

Full Solution Packet

Operator-valued Schur classes satisfy Sun's Wiener lemma

Status: candidate full result, likely valid

Target: Remark 3.11(2) of arXiv:2407.16416

Source and exact future-work problem

The source is Köhldorfer–Balazs, *Wiener pairs of Banach algebras of operator-valued matrices*, arXiv:2407.16416, published in the *Journal of Mathematical Analysis and Applications* 549 (2025), 129525 [1]. On page 18, Remark 3.11(2) says:

However, since ν is weakly growing by assumption, we have $\mathcal{S}_\nu^1 \subseteq \mathcal{S}_{\nu_\delta}^1$. Therefore, Theorem 3.7 implies that

$$r_{\mathcal{S}_\nu^1}(A) \leq r_{\mathcal{S}^1}(A) = r_{\mathcal{B}(\ell^2(X; \mathcal{H}))}(A) \quad (\forall A = A^* \in \mathcal{S}_\nu^1).$$

Conversely, by (2.2), we have $r_{\mathcal{B}(\ell^2(X; \mathcal{H}))}(A) \leq r_{\mathcal{S}_\nu^1}(A)$ for all $A = A^* \in \mathcal{S}_\nu^1$ since $\mathcal{S}_\nu^1 \subseteq \mathcal{B}(\ell^2(X; \mathcal{H}))$. Thus (3.20) follows. Now, having established the desired equation of spectral radii, the rest of the theorem follows from Hulanicki's lemma (Proposition 2.1). $\square$

Remark 3.11. 1.) The weak growth condition in Theorem 3.10 is essential. In fact, by giving a concrete counter-example, Tessera proved in [61] that the unweighted scalar-valued Schur algebra $\mathcal{S}^1(\mathbb{N})$ is not inverse-closed in $\mathcal{B}(\ell^p(\mathbb{N}; \mathbb{C}))$ for each $1 < p < \infty$. As emphasized in the same reference, $\mathcal{S}^1(\mathbb{N})$ is not even symmetric [43].

2.) Sun proved in [58] the inverse-closedness of the scalar-valued versions of $\mathcal{S}_\nu^p(X)$ in $\mathcal{B}(\ell^2(X; \mathbb{C}))$. We believe that the methods from [58] can be used to obtain analogous results in the $\mathcal{B}(\mathcal{H})$ -valued setting. We leave a detailed investigation to future research.

In text: Sun proved inverse-closedness of the scalar-valued versions of $\mathcal{S}_\nu^p(X)$ in $\mathcal{B}(\ell^2(X; \mathbb{C}))$. The authors expect analogous results for $\mathcal{B}(\mathcal{H})$ -valued matrices and leave the detailed investigation to future research.

We prove the operator-valued analogue of Sun's theorem under his original admissibility hypothesis: the weight is $(p, 2)$ -admissible. The local dimension of $\mathcal{H}$ is arbitrary, including infinite.

Definitions and hypotheses

Let (X, ρ) be a discrete pseudometric space whose counting measure has polynomial growth, and let $\mathcal{H}$ be a Hilbert space. For $1 \leq p \leq \infty$ and a symmetric weight w on $X \times X$, define $\mathcal{S}_w^p(X; \mathcal{B}(\mathcal{H}))$ to consist of block matrices $A = (A_{xy})_{x, y \in X}$ with $A_{xy} \in \mathcal{B}(\mathcal{H})$ and

$$\|A\|_{p, w} := \max \left\{ \sup_x \left\| \left(\|A_{xy}\| w(x, y) \right)_y \right\|_{\ell^p}, \sup_y \left\| \left(\|A_{xy}\| w(x, y) \right)_x \right\|_{\ell^p} \right\} < \infty.$$

For $p = \infty$ this has the usual supremum interpretation. This is the operator-valued space $\mathcal{S}_\nu^p(X)$ in [1] when $w(x, y) = \nu(x - y)$.

We use Sun's definition of a (p, r) -admissible weight [2, Definition in Section 2.2]. Thus there are a companion weight v , constants $D > 0$ and $0 < \theta < 1$, and, writing p' and r' for conjugate exponents, the following properties:

$$w(x, y) \leq D(w(x, z)v(z, y) + v(x, z)w(z, y)), \quad (1)$$

$$\sup_x \|(v/w)(x, \cdot)\|_{p'} + \sup_y \|(v/w)(\cdot, y)\|_{p'} < \infty, \quad (2)$$

$$\inf_{\tau > 0} (a_{r'}(\tau) + b_{p'}(\tau)t) \leq Dt^\theta \quad (t \geq 1). \quad (3)$$

Here $a_{r'}(\tau)$ is the uniform $\ell^{r'}$ size of v on the ρ -ball of radius τ , in rows and columns, while $b_{p'}(\tau)$ is the uniform $\ell^{p'}$ size of v/w outside that ball, again in rows and columns. These are exactly Sun's formulas (2.8)–(2.12); we have recalled their content to make clear where the proof uses the hypothesis.

Sun notes immediately after the definition that (p, r) -admissibility implies $(p, \tilde{r})$ -admissibility whenever $r \leq \tilde{r}$. The single crucial move below is therefore legitimate:

$$(p, 2)\text{-admissible} \implies (p, \infty)\text{-admissible}. \quad (4)$$

Proof intuition

Sun's scalar proof controls a local row/column norm of a matrix power by an ambient operator norm. At exponent $r = 2$, its delta-vector argument is false for operator blocks: many norm-one blocks in one row may act on mutually orthogonal directions, so their scalar ℓ^2 norm can be large although the block-row operator has norm one.

The admissibility hypothesis contains the repair. Run the same interpolation with auxiliary exponent $r = \infty$. Every individual block is a coordinate compression of the global operator, hence

$$\sup_{x,y} \|T_{xy}\| \leq \|T\|_{\mathcal{B}(\ell^2(X; \mathcal{H}))}.$$

This estimate is valid for arbitrary block size. The resulting interpolation inequality makes $\mathcal{S}_w^p$ a differential subalgebra of the ambient operator algebra. A dyadic spectral-radius argument and Hulanicki's lemma then give inverse-closedness.

The differential estimate

Set

$$\mathfrak{B} = \mathcal{B}(\ell^2(X; \mathcal{H})).$$

Lemma 1 (Operator-valued Sun estimate). *Assume that w is $(p, 2)$ -admissible. Then $\mathcal{S}_w^p(X; \mathcal{B}(\mathcal{H}))$ is a unital Banach $*$ -algebra continuously embedded in $\mathfrak{B}$. Moreover, for some constants $C > 0$ and $0 < \theta < 1$ depending only on the admissibility data,*

$$\|T^2\|_{p,w} \leq C \|T\|_{p,w}^{1+\theta} \|T\|_{\mathfrak{B}}^{1-\theta} \quad (T \in \mathcal{S}_w^p). \quad (5)$$

Proof. Completeness of the operator-valued space is Proposition 3.2 of [1]. We prove the algebra, embedding, and differential estimates.

Matrix multiplication and (1), followed by the standard weighted Jensen–Schur estimate for composing an ℓ^p row/column matrix with an ℓ^1 row/column matrix, give the compensated product estimate

$$\|RS\|_{p,w} \leq C_0(\|R\|_{p,w} \|S\|_{1,v} + \|R\|_{1,v} \|S\|_{p,w}). \quad (6)$$

This is Sun’s estimate (3.3). It remains valid for operator blocks because the only entrywise input is

$$\|(RS)_{xy}\| \leq \sum_z \|R_{xz}\| \|S_{zy}\|.$$

No commutativity is used.

By Hölder’s inequality and (2),

$$\|T\|_{1,v} \leq C_1 \|T\|_{p,w}. \quad (7)$$

Since $v \geq 1$, the operator-valued Schur test then yields

$$\|T\|_{\mathfrak{B}} \leq C_1 \|T\|_{p,w}. \quad (8)$$

Equations (6) and (7) prove the Banach algebra property. Symmetry of w gives $\|T^*\|_{p,w} = \|T\|_{p,w}$, and the diagonal identity belongs to the algebra because X has counting measure and $w(x, x)$ is uniformly bounded.

We now use (p, ∞) -admissibility from (4). Split each row and column of T into $\rho \leq \tau$ and $\rho > \tau$. On the near part, use the block-compression estimate

$$\|T_{xy}\| = \|P_x T P_y\| \leq \|T\|_{\mathfrak{B}}, \quad (9)$$

where P_x is the coordinate projection onto the x -copy of $\mathcal{H}$. On the far part, use Hölder’s inequality with w and v/w . Taking the row and column suprema gives

$$\|T\|_{1,v} \leq a_1(\tau) \|T\|_{\mathfrak{B}} + b_{p'}(\tau) \|T\|_{p,w}. \quad (10)$$

Let $K = \max\{1, C_1\}$ in (8) and put $t = K \|T\|_{p,w} / \|T\|_{\mathfrak{B}}$ when $T \neq 0$. Then $t \geq 1$. Take the infimum in (10) and apply (3); enlarging constants to absorb K gives

$$\|T\|_{1,v} \leq C_2 \|T\|_{\mathfrak{B}}^{1-\theta} \|T\|_{p,w}^{\theta}. \quad (11)$$

The zero operator is trivial. Finally apply (6) with $R = S = T$ and then (11). This yields (5). $\square$

Main result

Theorem 1 (Operator-valued Wiener lemma). *Let $1 \leq p \leq \infty$, let (X, ρ) be a discrete pseudometric space with counting measure of polynomial growth, and let w be a $(p, 2)$ -admissible weight in the sense of Sun. For every Hilbert space $\mathcal{H}$,*

$$\mathcal{S}_w^p(X; \mathcal{B}(\mathcal{H})) \subseteq \mathcal{B}(\ell^2(X; \mathcal{H}))$$

is inverse-closed. Equivalently, if $A \in \mathcal{S}_w^p(X; \mathcal{B}(\mathcal{H}))$ is invertible on $\ell^2(X; \mathcal{H})$, then

$$A^{-1} \in \mathcal{S}_w^p(X; \mathcal{B}(\mathcal{H})).$$

The algebra $\mathcal{S}_w^p(X; \mathcal{B}(\mathcal{H}))$ is also symmetric.

Proof. Write $\mathcal{A} = \mathcal{S}_w^p(X; \mathcal{B}(\mathcal{H}))$. The continuous unital inclusion $\mathcal{A} \hookrightarrow \mathfrak{B}$ always gives

$$r_{\mathfrak{B}}(T) \leq r_{\mathcal{A}}(T). \quad (12)$$

We prove the reverse inequality.

Apply Lemma 1 to T^{2^n} . With

$$A_n = \|T^{2^n}\|_{p,w}^{1/2^n}, \quad B_n = \|T^{2^n}\|_{\mathfrak{B}}^{1/2^n}, \quad \alpha = \frac{1+\theta}{2} < 1,$$

we obtain

$$A_{n+1} \leq C^{1/2^{n+1}} A_n^\alpha B_n^{1-\alpha}. \quad (13)$$

The sequence (A_n) is bounded (equivalently, replace the algebra norm by an equivalent submultiplicative norm), and Gelfand's formula gives $B_n \rightarrow r_{\mathfrak{B}}(T)$. If $\lambda > r_{\mathfrak{B}}(T)$, then $B_n \leq \lambda$ for all large n . Taking the limsup in (13) yields

$$L \leq L^\alpha \lambda^{1-\alpha}, \quad L := \limsup_n A_n.$$

Thus either $L = 0$ or $L \leq \lambda$. Letting $\lambda \downarrow r_{\mathfrak{B}}(T)$ gives

$$r_{\mathcal{A}}(T) \leq r_{\mathfrak{B}}(T).$$

Together with (12), the spectral radii agree (in particular, they agree for every self-adjoint $T \in \mathcal{A}$).

Hulanicki's lemma for a unital Banach $*$ -subalgebra of a C^* -algebra now implies that $\mathcal{A}$ is inverse-closed in $\mathfrak{B}$ and is symmetric; see [1, Proposition 2.1]. This proves the theorem. $\square$

Corollary 1 (Polynomial off-diagonal weights). *Let*

$$d_X = \inf\{d : \#B_\rho(x, R) \leq CR^d \text{ uniformly in } x, R \geq 1\}.$$

For $s > d_X(1 - 1/p)$, put $w_s(x, y) = (1 + \rho(x, y))^s$. Then $\mathcal{S}_{w_s}^p(X; \mathcal{B}(\mathcal{H}))$ is inverse-closed in $\mathcal{B}(\ell^2(X; \mathcal{H}))$. In particular, for a relatively separated $X \subset \mathbb{R}^d$, the sufficient condition $s > d(1 - 1/p)$ gives the operator-valued extension requested in Remark 3.11(2) of [1].

Proof. Sun's Example A.2 proves that w_s is (p, r) -admissible under the displayed condition, hence in particular $(p, 2)$ -admissible. Apply Theorem 1. $\square$

The same argument applies to the subexponential weights in Sun's Example A.3 (under its metric-space hypothesis).

Verification notes

The proof was audited at the four places where scalar-to-block extensions can fail.

- The entrywise product estimate uses only $\|RS\| \leq \|R\| \|S\|$ in $\mathcal{B}(\mathcal{H})$, so noncommutativity is harmless.
- The scalar $r = 2$ delta-vector estimate was *not* imported. It is replaced by $r = \infty$ and the exact compression inequality (9).

- Sun explicitly records the monotonic implication $(p, 2)$ -admissible $\Rightarrow (p, \infty)$ -admissible, so the changed auxiliary exponent does not strengthen the theorem’s hypothesis.
- The passage from the differential estimate to inverse-closedness is the dyadic spectral-radius argument above followed by the same Hulanicki lemma used in the source paper.

No computational or finite-dimensional test is used as evidence. The proof is uniform in $\dim \mathcal{H}$.

Novelty check and limitations

Before promotion, the run indexes were searched for arXiv:2407.16416, “Wiener pairs”, “operator-valued Schur”, “inverse-closed”, and “ $\mathcal{S}_w^p$ ”. No duplicate solution or attempt was found.

A bounded web search on 2026-07-19 used the exact source title and phrases combining “operator-valued”, “Schur class”, “Wiener lemma”, and “inverse-closed”. It found the 2025 journal version of the source and the authors’ arXiv:2501.09603 follow-up on the operator-valued Jaffard algebra, but no paper stating the $\mathcal{S}_w^p$ extension proved here. The search also checked Sun’s original 2007 paper and author-hosted PDF. Novelty is therefore plausible, not certified.

The theorem uses exactly the $(p, 2)$ -admissibility hypothesis of Sun’s scalar result. It does not claim inverse-closedness for arbitrary weights for which $\mathcal{S}_w^p$ is merely defined. This matches the natural precise reading of the source’s requested “analogous results.”

Human review recommendation

Send the packet to an expert in inverse-closed matrix algebras. The most important review point is the interpolation step (10)–(11): verify the convention for Sun’s $a_{r'}$ and $b_{p'}$ and the monotonic passage from $r = 2$ to $r = \infty$. After that, the block-compression and spectral-radius arguments are dimension-free and elementary.

References

- [1] Lukas Köhldorfer and Peter Balazs, *Wiener pairs of Banach algebras of operator-valued matrices*, J. Math. Anal. Appl. 549 (2025), 129525; arXiv:2407.16416.
- [2] Qiyu Sun, *Wiener’s lemma for infinite matrices*, Trans. Amer. Math. Soc. 359 (2007), 3099–3123.